1 Logistic Regression I

An article about the contributions of star players in the National Basketball Association (by M. L. Jones and R. J. Parker, Chance: 23, 39-45, 2010) reported prediction equations for the probability π of a win in a game for a player, using as predictors

- `ortg` = player's offensive rating in the game, which is the number of points produced per hundred possessions,
- `drtg` = player's defensive rating in the game, which is the number of points allowed per hundred possessions (the lower the better)
- `home` which indicates whether the game was played at home (1 = yes, 0 = no).

For LeBron James using data from the 2008-2009 season.

$$\text{logit}(\hat{\pi}) = 1.379 + 0.119(\text{ortg}) - 0.139(\text{drtg}) + 3.393(\text{home})$$

1.1 Quartiles

Over the season, James's quartiles (lower, median, upper) were (108.7,123.2, 136.1) for `ortg` and (91.7, 99.5, 107.7) for `drtg`. Summarize the `ortg` effect for James by comparing `ortg` at its upper and lower quartiles. Do this at the median level of `drtg`, separately for home and away games. Repeat for the `drtg` effect, and compare.

```
ortg <- list("lower" = 108.7, "median" = 123.2, "upper" = 136.1)
drtg <- list("lower" = 91.7, "median" = 99.5, "upper" = 107.7)
home <- list("yes" = 1, "no" = 0)

get_probability <- function(ortg, drtg, home){
  logit <- 1.379 + 0.119*ortg - 0.139*drtg + 3.393*home
  return(exp(logit)/(1 + exp(logit)))
}

median_drtg_combinations <- expand.grid(
  ortg = c(ortg$lower, ortg$upper), home = c(home$yes, home$no),
  drtg = drtg$median
) %>% arrange(home, ortg)

median_ortg_combinations <- expand.grid(
  ortg = ortg$median, home = c(home$yes, home$no),
  drtg = c(drtg$lower, drtg$upper)
) %>% arrange(home, drtg)

input <- rbind(median_drtg_combinations, median_ortg_combinations)

input$win_prob <- apply(input, 1, function(row){
  get_probability(ortg = row['ortg'], drtg = row['drtg'], home = row['home'])})
```

Table 1.1: Table of Results

ortg	home	drtg	$\hat{\pi}$
108.7	0	99.5	0.61864
136.1	0	99.5	0.97690
108.7	1	99.5	0.97970
136.1	1	99.5	0.99921
123.2	0	91.7	0.96421
123.2	0	107.7	0.74450
123.2	1	91.7	0.99875
123.2	1	107.7	0.98860

When away from home, LeBron's win probability given the same `drtg` can jump from 0.61864 to 0.9769. The difference is much smaller when game is held in his home, favoring the player. On the other hand, LeBron's win probability given the same `ortg` decreases from 0.96421 to 0.7445 when game is away from his home and a smaller difference is noticeable when the game is within his home because it's almost 100%. This implies that LeBron's win is influenced hugely by where the game is held.

1.2 Home effect

Summarize the home effect by

comparing $\hat{\pi}$ for home and away games, at the median levels of `ortg` and `drtg`,

```
win_probs <- lapply(home, function(x)get_probability(ortg = ortg$median,
                                                     drtg = drtg$median,
                                                     home = x))
```

LeBron's win probability is 0.99632 is higher when game is held in his home and is smaller otherwise (0.90108).

Interpreting its coefficient in the fitted logistic equation.

Aside:

$$\exp(\text{logit}[\pi(x_3)]) = \exp\left(\log \frac{\pi(x_3)}{1-\pi(x_3)}\right) = \exp(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3)$$

$$\longleftrightarrow \frac{\pi(x_3)}{1-\pi(x_3)} = \exp(\beta_0)\exp(\beta_1 x_1)\exp(\beta_2 x_2)\exp(\beta_3 x_3) \qquad \text{odds at } X_3 = x_3$$

$$\begin{aligned}
\frac{\pi(x_3+1)}{1-\pi(x_3+1)} &= \exp(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + (\beta_3 x_3 + 1)) \\
&= \exp(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_3) \\
&= \exp(\beta_0)\exp(\beta_1 x_1)\exp(\beta_2 x_2)\exp(\beta_3 x_3)\exp(\beta_3) \\
&= \frac{\pi(x_3)}{1-\pi(x_3)}\exp(\beta_3) \qquad \text{substitute odds at } X_3 = x_3
\end{aligned} \tag{1.1}$$

Coefficient of `home`:

- A game held in LeBron's `home` multiplies the odds of winning by $\exp(3.393) = 29.75508$, holding `ortg` and `drtg` constant.

2 Logistic Regression II

For a study using logistic regression to determine characteristics associated with remission in cancer patients, the table below shows the most important explanatory variable, a labeling index (`LI`) that measures proliferative activity of cells after a patient receives an injection of tritiated thymidine. It represents the percentage of cells that are "labeled." The response measured whether the patient achieved remission. Software reports a logistic regression model using `LI` to estimate $\pi = P(\text{remission})$.

LI	Number of Cases	Number of Remissions	LI	Number of Cases	Number of Remissions	LI	Number of Cases	Number of Remissions
8	2	0	18	1	1	28	1	1
10	2	0	20	3	2	32	1	0
12	3	0	22	2	1	34	1	1
14	3	0	24	1	0	38	3	2
16	3	0	26	1	1			

Source: Data reprinted with permission from E. T. Lee, *Comput. Prog. Biomed.* **4**: 80–92, 1974.

```
                          Intercept   Intercept and
            Criterion       Only        Covariates
            −2 Log L      34.372         26.073

Parameter   Estimate   Standard Error   Chi-Square   Pr > ChiSq
Intercept    −3.7771       1.3786         7.5064       0.0061
li            0.1449       0.0593         5.9594       0.0146

                    Odds Ratio Estimates
    Effect   Point Estimate   95% Wald Confidence Limits
    li           1.156        1.029          1.298

                 Estimated Covariance Matrix
            Variable    Intercept      li
            Intercept    1.900616   −0.07653
            li          −0.07653     0.003521

      Obs   li   remiss   n   pi_hat     lower     upper
       1    8     0      2   0.06797   0.01121   0.31925
       2   10     0      2   0.08879   0.01809   0.34010
```

2.1 Software results

Show how software obtained A = 0.068 when LI = 8.

```
slope <- 0.1449; intercept <- -3.7771; se <- 0.0593

get_li_prob <- function(li){
  ex <- exp(intercept+slope*li)
  return(ex/(1+ex))
  }
```

$$\hat{\pi}(x) = \frac{\exp(-3.7771 + 0.1449\text{li})}{1 + \exp(-3.7771 + 0.1449\text{li})} = 0.068$$

2.2 A given LI

Show that A = 0.50 when LI = 26.0.

Using `get_li_prob`, we get 0.49758.

2.3 RoC in A

Show that the rate of change in A is 0.009 when LI = 8 and 0.036 when LI= 26.

```
get_roc <- function(li){
  return(slope * get_li_prob(li = li) * (1 - get_li_prob(li = li)))
}
```

$$\begin{aligned}
\frac{\partial \pi(x)}{\partial x} &= \frac{\partial}{\partial x} \frac{\exp(\alpha + \beta x)}{1 + \exp(\alpha + \beta x)} \\
&= \frac{(1 + \exp(\alpha + \beta x)) \frac{\partial}{\partial x} \exp(\alpha + \beta x) - \exp(\alpha + \beta x) \frac{\partial}{\partial x}(1 + \exp(\alpha + \beta x))}{(1 + \exp(\alpha + \beta x))^2} \\
&= \frac{(1 + \exp(\alpha + \beta x))\beta \exp(\alpha + \beta x) - \exp(\alpha + \beta x)\beta \exp(\alpha + \beta x)}{(1 + \exp(\alpha + \beta x))^2} \\
&= \frac{\beta \exp(\alpha + \beta x) + \beta \exp(\alpha + \beta x)^2 - \beta \exp(\alpha + \beta x)^2}{(1 + \exp(\alpha + \beta x))^2} \\
&= \beta \frac{\exp(\alpha + \beta x)}{1 + \exp(\alpha + \beta x)} \frac{1}{1 + \exp(\alpha + \beta x)} \\
&= \beta \pi(x)(1 - \pi(x)) \qquad (2.1)
\end{aligned}$$

We use equation (2.1) and `get_roc` to compute the rate of change when $LI = 8$: 0.00918. When $LI = 26$, rate of change is 0.03622

2.4 Change in A

The lower quartile and upper quartile for LI are 14 and 28. Show that A increases by 0.42, from 0.15 to 0.57, between those values.

`get_li_prob(li = 28) - get_li_prob(li = 14)`= 0.42133

2.5 Estimated odds of remission

For a unit increase in LI, show that the estimated odds of remission multiply by 1.16.

```
get_odds_factor <- function(li){
  num <- get_li_prob(li+1) / (1 - get_li_prob(li+1))
  den <- get_li_prob(li) / (1 - get_li_prob(li))
  return(num/den)
}
```

Applying the result from Equation and using any `li` value for `get_odds_factor`, we obtain 1.15592

2.6 CI for the odds ratio

Explain how to obtain the confidence interval reported for the odds ratio. Interpret.

For every unit increase in li, the odds of remission gets multiplied by a quantity from 1.02909 to 1.29839. Align with this, both `LB` and `UB` are > 1.

2.7 Wald test

Construct a Wald test for the effect. Interpret.

Let $H_0 : \beta = 0$ and use Wald test statistic.

$$\frac{(\hat{\beta}_1 - \beta_1)^2}{Var(\hat{\beta}_1)} = \frac{(0.1449 - 0)^2}{0.0593^2} = 5.97073 > 3.8416 = \chi_1^2$$

We are 95% confident that if we repeatedly collect sample and estimate β, its value will be found within the lower and upper bounds, and this does not include 0. We reject independence hypothesis.

2.8 LRT

Conduct a likelihood-ratio test for the effect, showing how to construct the test statistic using the ^-2logL values reported.

The null deviance assumes an intercept only model will the residual deviance considers all the predictors included in the model. In this case, it's just LI. The df for the null deviance is 26. We have 27 rows in the long form of the data and we only estimate parameter which is the

intercept which make the $df = 27 - 1 = 25$. For the residual deviance, we estimate the slope and the intercept so we have two parameters giving us a $df = 27 - 2 = 25$

Let H_0 : The models are equally good (The simpler model fits the data equally as good as the complex one) and use LRT.

```
dataset$Number.of.NonRemission <- dataset$Number.of.Cases -
  dataset$Number.of.Remissions

nonremission <- dataset %>%
  dplyr::select(`LI`, `Number.of.NonRemission`)
nonremission$remission <- sapply(dataset$Number.of.NonRemission,
                                 function(x){rep(0, x)})
remission <- dataset %>%
  dplyr::select(`LI`, `Number.of.Remissions`)
remission$remission <- sapply(dataset$Number.of.Remissions,
                              function(x){rep(1, x)})

prepared_dataset <- rbind(
  nonremission %>% unnest(remission) %>% dplyr::select(LI, remission),
  remission %>% unnest(remission) %>% dplyr::select(LI, remission)
)
```

```
fit <- glm(remission~LI,data=prepared_dataset,family=binomial(link="logit"))
```

```
Call:
glm(formula = remission ~ LI, family = binomial(link = "logit"),
   data = prepared_dataset)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -3.7771     1.3786   -2.74   0.0061 **
LI            0.1449     0.0593    2.44   0.0146 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

   Null deviance: 34.372  on 26  degrees of freedom
Residual deviance: 26.073  on 25  degrees of freedom
AIC: 30.07

Number of Fisher Scoring iterations: 4
```

`R` defaults to `link = "logit"` for `family = binomial` so the `link` may be removed safely. Note this is another way to specify the model without having to convert to long form:

```
grouped_fit_1 <- glm(
  cbind(Number.of.Remissions, Number.of.NonRemission) ~ LI,
  data = dataset, family = binomial(link="logit"))
```

Another way to specify the model using proportion and weight.

```r
grouped_fit_2 <- glm(
  Number.of.Remissions/Number.of.Cases ~ LI,
  data = dataset, weights = Number.of.Cases, family = binomial(link="logit"))
```

For the likelihood ratio statistic, we use G^2

$$
\begin{aligned}
G^2 &= 2\log\left(\frac{L_{\text{saturated}}}{L_{\text{nested}}}\right) \\
&= 2\log(L_{\text{saturated}}) - 2\log(L_{\text{reduced}}) \\
&= -1(26.07296) - [-1(34.37177)] \\
&= 34.37177 - 26.07296 \\
&= 8.2988
\end{aligned} \tag{2.2}
$$

Since $G^2 = 8.2988 > 3.8416 = \chi^2_{26-25}$, reject the H_0 and prefer the model that includes `LI`.

3 Logistic Regression III

Hastie and Tibshirani (1990, p. 282) described a study to determine risk factors for kyphosis, severe forward flexion of the spine following corrective spinal surgery. The age in months at the time of the operation for the 18 subjects for whom kyphosis was present were 12, 15, 42, 52, 59, 73, 82, 91, 96, 105, 114, 120, 121, 128, 130, 139, 139, 157 and for 22 of the subjects for whom kyphosis was absent were 1, 1,2, 8, 11, 18,22,31,37,61,72,81,97, 112, 118, 127, 131, 140, 151, 159, 177,206.

3.1 Model fitting

Fit a logistic regression model using age as a predictor of whether kyphosis is present. Test whether age has a significant effect.

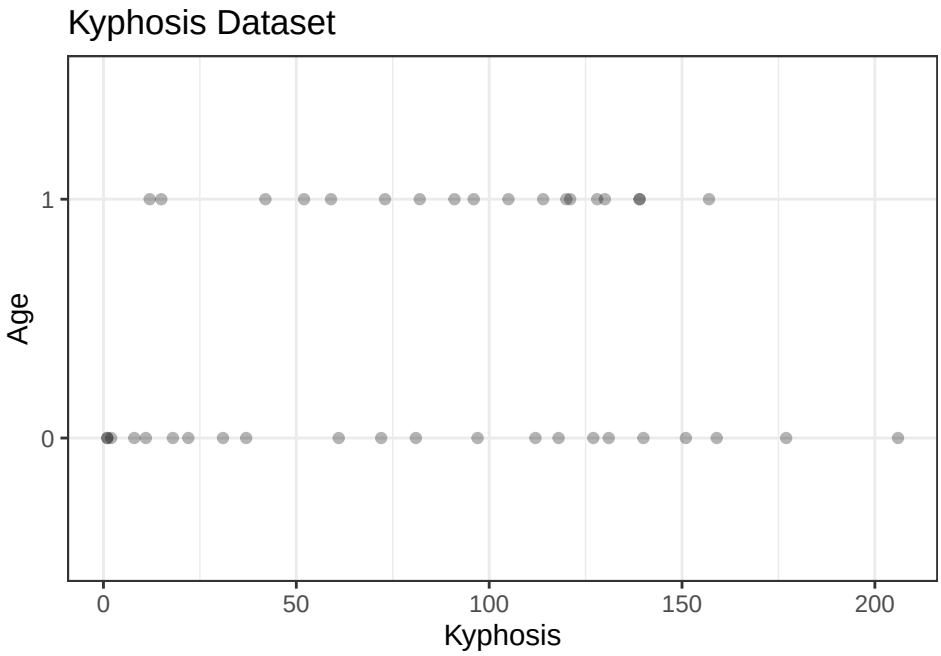
```r
kyphosis_fit1 <- glm(kyphosis~age,data=kyphosis_dataset,
                     family=binomial(link="logit"))
```

```
Call:
glm(formula = kyphosis ~ age, family = binomial(link = "logit"),
   data = kyphosis_dataset)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -0.57269    0.60239   -0.95     0.34
age          0.00430    0.00585    0.73     0.46

(Dispersion parameter for binomial family taken to be 1)

   Null deviance: 55.051  on 39  degrees of freedom
Residual deviance: 54.504  on 38  degrees of freedom
AIC: 58.5

Number of Fisher Scoring iterations: 4
```

Compute `kyphosis_fit1$null.deviance - kyphosis_fit1$deviance` $= 0.54689 < 3.8416 = \chi^2_1$. We do not reject the H_0. For `kyphosis_fit1`, as shown by the very small value of G^2, `age` does not have a significant effect.

3.2 Data plotting

Plot the data. Note the difference in dispersion on age at the two levels of kyphosis. Fit the model $\text{logit}(\pi) = \alpha + \beta_1 x + \beta_2 x^2$. Test the significance of the squared age term, plot the fit, and interpret.




```r
kyphosis_fit2 <- glm(kyphosis~age + I(age^2),data=kyphosis_dataset,
                     family=binomial(link="logit"))
```

```
Call:
glm(formula = kyphosis ~ age + I(age^2), family = binomial(link = "logit"),
   data = kyphosis_dataset)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -2.046255   0.994348   -2.06    0.040 *
age          0.060040   0.026781    2.24    0.025 *
I(age^2)    -0.000328   0.000156   -2.10    0.036 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

   Null deviance: 55.051  on 39  degrees of freedom
Residual deviance: 48.228  on 37  degrees of freedom
AIC: 54.23

Number of Fisher Scoring iterations: 4
```

Let H_0 : The models are equally good (The simpler model fits the data equally as good as the complex one) and use LRT.

Compute `kyphosis_fit1$deviance - kyphosis_fit2$deviance` $= 6.27623 > 3.8416 = \chi^2_1$. We reject the null hypothesis and prefer the model that includes the squared age term.

Interpretation:

$$
\begin{aligned}
\frac{\pi(x)}{1-\pi(x)} &= \exp(\alpha + \beta_1 x + \beta_2 x^2) \\
\frac{\pi(x+1)}{1-\pi(x+1)} &= \exp(\alpha + \beta_1(x+1) + \beta_2(x+1)^2) \\
&= \exp(\alpha + \beta_1 x + \beta_1 + \beta_2(x^2 + 2x + 1)) \\
&= \exp(\alpha + \beta_1 x + \beta_1 + \beta_2 x^2 + 2\beta_2 x + \beta_2)) \\
&= \exp(\alpha + \beta_1 x + \beta_2 x^2 + 2\beta_2 x + \beta_2 + \beta_1)) \\
&= \exp(\alpha + \beta_1 x + \beta_2 x^2)\exp(2\beta_2 x + \beta_2 + \beta_1)) \\
&= \frac{\pi(x)}{1-\pi(x)}\exp(2\beta_2 x + \beta_2 + \beta_1)) \\
&= \frac{\pi(x)}{1-\pi(x)}\exp(2\beta_2 x)\exp(\beta_2)\exp(\beta_1)
\end{aligned}
$$

For every increase in months of age, the odds of getting severe forward flexion of the spine following corrective spinal surgery is multiplied by `exp(2` $\times$ -0.00033$\times$ `age)` $\times$ `exp(` -0.00033) $\times$ `exp(` 0.06004) `= exp(` -0.00066 $\times$ `age)` $\times$ 1.06153

4 Logistic Regression IV

For the table below, treating marijuana use as the response variable, build a model with alcohol use, cigarette use, gender, and race as potential explanatory variables. Summarize your strategy for selecting a model and interpret your final choice of model.

		Marijuana Use							
		Race = White				Race = Other			
		Female		Male		Female		Male	
Alcohol Use	Cigarette Use	Yes	No	Yes	No	Yes	No	Yes	No
Yes	Yes	405	268	453	228	23	23	30	19
	No	13	218	28	201	2	19	1	18
No	Yes	1	17	1	17	0	1	1	8
	No	1	117	1	133	0	12	0	17

Source: Harry Khamis, Wright State University.

Prepare the dataset:

```
marijuana_dataset <- data.frame(
  rbind(c(1, 1, 1, 1, 405, 268), c(1, 0, 1, 1, 13, 218), c(0, 1, 1, 1, 1, 17),
```

```
        c(0, 0, 1, 1, 1, 117), c(1, 1, 0, 1, 453, 228), c(1, 0, 0, 1, 28, 201),
        c(0, 1, 0, 1, 1, 17), c(0, 0, 0, 1, 1, 133), c(1, 1, 1, 0, 23, 23),
        c(1, 0, 1, 0, 2, 19), c(0, 1, 1, 0, 0, 1), c(0, 0, 1, 0, 0, 12),
        c(1, 1, 0, 0, 30, 19), c(1, 0, 0, 0, 1, 18), c(0, 1, 0, 0, 1, 8),
        c(0, 0, 0, 0, 0, 17)))

colnames(marijuana_dataset) = c("Alcohol Use", "Cigarette Use", "Female",
                                "White", "Marijuana Use 1", "Marijuana Use 0")

marijuana_dataset <- marijuana_dataset %>%
  mutate(
    `Alcohol Use` = factor(`Alcohol Use`),
    `Cigarette Use` = factor(`Cigarette Use`),
    Female = factor(Female), White = factor(White)
    )
```

The saturated model includes interaction of all four variables. We use the `step` function to administer backward elimination. By default, `R` uses `AIC` and stops dropping variables once the drop leads to increase in `AIC`.

```
model_4way <- glm(
  cbind(`Marijuana Use 1`, `Marijuana Use 0`) ~
    `Alcohol Use` * `Cigarette Use` * Female * White,
  data = marijuana_dataset, family = binomial
)

backward_elimination_model <-  step(model_4way, direction = "backward",
                                    trace = 0)

step_results <- backward_elimination_model$anova # summary(backward_model)
```

Table 4.1: Backward Elimination Steps using ‘step‘ function in ‘R‘

Step	Df	Deviance	Resid. Df	Resid. Dev	AIC
	NA	NA	0	0.00000	77.996
- ‘Alcohol Use‘:‘Cigarette Use‘:Female:White	1	0.00000	1	0.00000	75.996
- ‘Alcohol Use‘:‘Cigarette Use‘:Female	1	0.11529	2	0.11529	74.111
- ‘Alcohol Use‘:Female:White	1	0.24368	3	0.35897	72.355
- ‘Alcohol Use‘:Female	1	0.25425	4	0.61322	70.609
- ‘Alcohol Use‘:‘Cigarette Use‘:White	1	0.54307	5	1.15629	69.152
- ‘Alcohol Use‘:White	1	0.09904	6	1.25534	67.251
- ‘Alcohol Use‘:‘Cigarette Use‘	1	0.35266	7	1.60800	65.604
- ‘Cigarette Use‘:Female:White	1	1.56949	8	3.17749	65.173
- ‘Cigarette Use‘:White	1	0.00825	9	3.18574	63.181
- Female:White	1	0.01522	10	3.20095	61.197
- ‘Cigarette Use‘:Female	1	1.49291	11	4.69386	60.689

Model selected from the automated backward elimination using `step` function:

```
Call:
glm(formula = cbind(`Marijuana Use 1`, `Marijuana Use 0`) ~ `Alcohol Use` +
   `Cigarette Use` + Female + White, family = binomial, data = marijuana_dataset)

Coefficients:
                Estimate Std. Error z value Pr(>|z|)
(Intercept)       -5.433      0.506  -10.74  < 2e-16 ***
`Alcohol Use`1     2.987      0.466    6.42  1.4e-10 ***
`Cigarette Use`1   2.859      0.164   17.40  < 2e-16 ***
Female1           -0.330      0.103   -3.21   0.0013 **
White1             0.299      0.201    1.48   0.1380
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

   Null deviance: 860.6086  on 15  degrees of freedom
Residual deviance:   4.6939  on 11  degrees of freedom
AIC: 60.69

Number of Fisher Scoring iterations: 5
```

To understand the entire process, we attempt to do it manually.

Let H_0 : Simpler model fits the data as well as the complex one.

(If G^2 is too small, do not reject. When difference is large, they are not performing equally well. Simple model is far from the performance of complex Simple deviates much from complex, reject and prefer complex.)

Drop the 4-way interaction term. The small value under LRT (G^2) shows that the term is not significant and does not add value so the simpler model is preferable.

```
Single term deletions

Model:
cbind(`Marijuana Use 1`, `Marijuana Use 0`) ~ `Alcohol Use` *
   `Cigarette Use` * Female * White
                                          Df Deviance AIC      LRT Pr(>Chi)
<none>                                        2.31e-10  78
`Alcohol Use`:`Cigarette Use`:Female:White  1 4.22e-10  76 1.92e-10        1
```

Continuing, we update the model by dropping the 4-way interaction term following the previous result. We then check what potentially happens if we drop any of the 3-way interaction terms. `LRT` shows that we can remove them. We start dropping the term with the highest p-value.

```
Single term deletions

Model:
cbind(`Marijuana Use 1`, `Marijuana Use 0`) ~ `Alcohol Use` +
   `Cigarette Use` + Female + White + `Alcohol Use`:`Cigarette Use` +
   `Alcohol Use`:Female + `Cigarette Use`:Female + `Alcohol Use`:White +
   `Cigarette Use`:White + Female:White + `Alcohol Use`:`Cigarette Use`:Female +
   `Alcohol Use`:`Cigarette Use`:White + `Alcohol Use`:Female:White +
   `Cigarette Use`:Female:White
                                     Df Deviance  AIC   LRT Pr(>Chi)
<none>                                    0.000 76.0
`Alcohol Use`:`Cigarette Use`:Female  1   0.115 74.1 0.115     0.73
```

```
‘Alcohol Use‘:‘Cigarette Use‘:White   1    0.218 74.2 0.218     0.64
‘Alcohol Use‘:Female:White            1    0.174 74.2 0.174     0.68
‘Cigarette Use‘:Female:White          1    1.549 75.5 1.549     0.21
```

We update the model again to remove the term with the highest p-value and see what happens if we drop more 3-way interactions.

```
[1] "......... Dropping ‘‘Alcohol Use‘:‘Cigarette Use‘:Female‘ ........."

Single term deletions

Model:
cbind(‘Marijuana Use 1‘, ‘Marijuana Use 0‘) ~ ‘Alcohol Use‘ +
   ‘Cigarette Use‘ + Female + White + ‘Alcohol Use‘:‘Cigarette Use‘ +
   ‘Alcohol Use‘:Female + ‘Cigarette Use‘:Female + ‘Alcohol Use‘:White +
   ‘Cigarette Use‘:White + Female:White + ‘Alcohol Use‘:‘Cigarette Use‘:White +
   ‘Alcohol Use‘:Female:White + ‘Cigarette Use‘:Female:White
                                  Df Deviance  AIC   LRT Pr(>Chi)
<none>                                  0.115 74.1
‘Alcohol Use‘:‘Cigarette Use‘:White  1    0.419 72.4 0.304     0.58
‘Alcohol Use‘:Female:White           1    0.359 72.4 0.244     0.62
‘Cigarette Use‘:Female:White         1    1.628 73.6 1.513     0.22
```

Repeating the same procedure of updating the model, looking at the values under LRT, and dropping the term with the highest p-value, we then reach the state where we have 2-way interaction terms.

```
[1] "......... Dropping ‘‘Alcohol Use‘:Female:White‘ ........."

Single term deletions

Model:
cbind(‘Marijuana Use 1‘, ‘Marijuana Use 0‘) ~ ‘Alcohol Use‘ +
   ‘Cigarette Use‘ + Female + White + ‘Alcohol Use‘:‘Cigarette Use‘ +
   ‘Alcohol Use‘:Female + ‘Cigarette Use‘:Female + ‘Alcohol Use‘:White +
   ‘Cigarette Use‘:White + Female:White + ‘Alcohol Use‘:‘Cigarette Use‘:White +
   ‘Cigarette Use‘:Female:White
                                  Df Deviance  AIC   LRT Pr(>Chi)
<none>                                  0.359 72.4
‘Alcohol Use‘:Female                 1    0.613 70.6 0.254     0.61
‘Alcohol Use‘:‘Cigarette Use‘:White  1    1.036 71.0 0.677     0.41
‘Cigarette Use‘:Female:White         1    1.895 71.9 1.537     0.22


[1] "......... Dropping ‘‘Alcohol Use‘:‘Cigarette Use‘:White‘ ........."


Single term deletions

Model:
cbind(‘Marijuana Use 1‘, ‘Marijuana Use 0‘) ~ ‘Alcohol Use‘ +
   ‘Cigarette Use‘ + Female + White + ‘Alcohol Use‘:‘Cigarette Use‘ +
   ‘Alcohol Use‘:Female + ‘Cigarette Use‘:Female + ‘Alcohol Use‘:White +
   ‘Cigarette Use‘:White + Female:White + ‘Cigarette Use‘:Female:White
                              Df Deviance  AIC   LRT Pr(>Chi)
<none>                                1.04 71.0
‘Alcohol Use‘:‘Cigarette Use‘  1     1.47 69.5 0.433     0.51
‘Alcohol Use‘:Female           1     1.16 69.2 0.120     0.73
‘Alcohol Use‘:White            1     1.17 69.2 0.135     0.71
‘Cigarette Use‘:Female:White   1     2.61 70.6 1.570     0.21
```

```
[1] "......... Dropping ``Cigarette Use`:Female:White` ........."


Single term deletions

Model:
cbind(`Marijuana Use 1`, `Marijuana Use 0`) ~ `Alcohol Use` +
   `Cigarette Use` + Female + White + `Alcohol Use`:`Cigarette Use` +
   `Alcohol Use`:Female + `Cigarette Use`:Female + `Alcohol Use`:White +
   `Cigarette Use`:White + Female:White
                             Df Deviance  AIC   LRT Pr(>Chi)
<none>                               2.61 70.6
`Alcohol Use`:`Cigarette Use`  1     3.02 69.0 0.413     0.52
`Alcohol Use`:Female           1     2.74 68.7 0.136     0.71
`Cigarette Use`:Female         1     4.16 70.2 1.549     0.21
`Alcohol Use`:White            1     2.76 68.8 0.158     0.69
`Cigarette Use`:White          1     2.61 68.6 0.001     0.97
Female:White                   1     2.62 68.6 0.011     0.92
```

We stick to the same procedure and arrive at such state where we only have the main effects. Notice that the automated process of backward elimination stopped in the presence of all variables in the main effects. However, LRT shows that `White` is not significant. We can try to further eliminate it.

```
[1] "......... Dropping ``Cigarette Use`:White` ........."

Single term deletions

Model:
cbind(`Marijuana Use 1`, `Marijuana Use 0`) ~ `Alcohol Use` +
   `Cigarette Use` + Female + White + `Alcohol Use`:`Cigarette Use` +
   `Alcohol Use`:Female + `Cigarette Use`:Female + `Alcohol Use`:White +
   Female:White
                             Df Deviance  AIC   LRT Pr(>Chi)
<none>                               2.61 68.6
`Alcohol Use`:`Cigarette Use`  1     3.03 67.0 0.417     0.52
`Alcohol Use`:Female           1     2.74 66.7 0.137     0.71
`Cigarette Use`:Female         1     4.16 68.2 1.549     0.21
`Alcohol Use`:White            1     2.77 66.8 0.162     0.69
Female:White                   1     2.62 66.6 0.012     0.91


[1] "......... Dropping `Female:White` ........."


Single term deletions

Model:
cbind(`Marijuana Use 1`, `Marijuana Use 0`) ~ `Alcohol Use` +
   `Cigarette Use` + Female + White + `Alcohol Use`:`Cigarette Use` +
   `Alcohol Use`:Female + `Cigarette Use`:Female + `Alcohol Use`:White
                             Df Deviance  AIC   LRT Pr(>Chi)
<none>                               2.62 66.6
`Alcohol Use`:`Cigarette Use`  1     3.03 65.0 0.415     0.52
`Alcohol Use`:Female           1     2.76 64.8 0.135     0.71
`Cigarette Use`:Female         1     4.17 66.2 1.547     0.21
`Alcohol Use`:White            1     2.79 64.8 0.171     0.68
```

```
[1] "......... Dropping ``Alcohol Use`:Female` ........."


Single term deletions

Model:
cbind(`Marijuana Use 1`, `Marijuana Use 0`) ~ `Alcohol Use` +
   `Cigarette Use` + Female + White + `Alcohol Use`:`Cigarette Use` +
   `Cigarette Use`:Female + `Alcohol Use`:White
                            Df Deviance  AIC   LRT Pr(>Chi)
<none>                              2.76 64.8
`Alcohol Use`:`Cigarette Use`  1     3.13 63.1 0.380     0.54
`Cigarette Use`:Female         1     4.22 64.2 1.464     0.23
`Alcohol Use`:White            1     2.87 62.9 0.117     0.73


[1] "......... Dropping ``Alcohol Use`:White` ........."


Single term deletions

Model:
cbind(`Marijuana Use 1`, `Marijuana Use 0`) ~ `Alcohol Use` +
   `Cigarette Use` + Female + White + `Alcohol Use`:`Cigarette Use` +
   `Cigarette Use`:Female
                            Df Deviance  AIC   LRT Pr(>Chi)
<none>                              2.87 62.9
White                          1     5.00 63.0 2.125     0.14
`Alcohol Use`:`Cigarette Use`  1     3.20 61.2 0.329     0.57
`Cigarette Use`:Female         1     4.33 62.3 1.461     0.23


[1] "......... Dropping ``Alcohol Use`:`Cigarette Use`` ........."


Single term deletions

Model:
cbind(`Marijuana Use 1`, `Marijuana Use 0`) ~ `Alcohol Use` +
   `Cigarette Use` + Female + White + `Cigarette Use`:Female
                      Df Deviance   AIC  LRT Pr(>Chi)
<none>                         3.2  61.2
`Alcohol Use`           1     94.3 150.3 91.1   <2e-16 ***
White                   1      5.4  61.4  2.2     0.14
`Cigarette Use`:Female  1      4.7  60.7  1.5     0.22
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1


[1] "......... Dropping ``Cigarette Use`:Female` ........."
```

```
Single term deletions

Model:
cbind(‘Marijuana Use 1‘, ‘Marijuana Use 0‘) ~ ‘Alcohol Use‘ +
   ‘Cigarette Use‘ + Female + White
               Df Deviance AIC LRT Pr(>Chi)
<none>                    5  61
‘Alcohol Use‘    1       95 149  91   <2e-16 ***
‘Cigarette Use‘  1      502 556 498   <2e-16 ***
Female           1       15  69  10   0.0013 **
White            1        7  61   2   0.1384
---
Signif. codes:  0 ’***’ 0.001 ’**’ 0.01 ’*’ 0.05 ’.’ 0.1 ’ ’ 1


[1] "..... Dropping ‘White‘ ....."

Single term deletions

Model:
cbind(‘Marijuana Use 1‘, ‘Marijuana Use 0‘) ~ ‘Alcohol Use‘ +
   ‘Cigarette Use‘ + Female
               Df Deviance AIC LRT Pr(>Chi)
<none>                    7  61
‘Alcohol Use‘    1      101 153  94   <2e-16 ***
‘Cigarette Use‘  1      505 557 498   <2e-16 ***
Female           1       17  69  10   0.0014 **
---
Signif. codes:  0 ’***’ 0.001 ’**’ 0.01 ’*’ 0.05 ’.’ 0.1 ’ ’ 1
```

Eliminating `White` left all the remaining variables significant. The reason why the automated process of backward elimination stopped with all the four variables present is because it has a lower AIC (60.68945) than the main effects model without `White` which has AIC 60.88463. We prefer the more parsimonious model since the difference in AIC is minimal.

Final model we select:

```
Call:
glm(formula = cbind(‘Marijuana Use 1‘, ‘Marijuana Use 0‘) ~ ‘Alcohol Use‘ +
   ‘Cigarette Use‘ + Female, family = binomial, data = marijuana_dataset)

Coefficients:
                Estimate Std. Error z value Pr(>|z|)
(Intercept)        -5.188      0.477  -10.88  < 2e-16 ***
‘Alcohol Use‘1      3.020      0.465    6.49  8.6e-11 ***
‘Cigarette Use‘1    2.859      0.164   17.41  < 2e-16 ***
Female1            -0.328      0.103   -3.20   0.0014 **
---
Signif. codes:  0 ’***’ 0.001 ’**’ 0.01 ’*’ 0.05 ’.’ 0.1 ’ ’ 1

(Dispersion parameter for binomial family taken to be 1)

   Null deviance: 860.609  on 15  degrees of freedom
Residual deviance:   6.889  on 12  degrees of freedom
AIC: 60.88

Number of Fisher Scoring iterations: 5
```

Model interpretation in terms of log-odds:

- The log-odds (logit) of using marijuana for someone who uses alcohol is 3.02015 higher than the log-odds for someone who does not use alcohol, holding other variables constant.
- The log-odds (logit) of using marijuana for someone who smokes cigarettes is 2.85913 higher than the log-odds for someone who does not smoke cigarettes, holding other variables constant.
- The log-odds (logit) of using marijuana for females is 0.32791 lower than the log-odds for males, holding other variables constant.
- The log-odds (logit) of using marijuana for males who do not smoke and do not drink alcohol is -5.18833.

Model interpretation in terms of odds: We take `exp` of the coefficients.

- The odds of using marijuana for someone who uses alcohol is 20.49445 times the odds for someone who does not use alcohol, holding other variables constant.
- The odds of using marijuana for someone who smokes cigarettes is 17.44634 times the odds for someone who does not smoke cigarettes, holding other variables constant.
- The odds of using marijuana for females is 0.72043 times the odds for males, holding other variables constant.
- The odds of using marijuana for males who do not smoke and do not drink alcohol is 0.00558.

5 Logistic Regression V

The table below shows the results of a study about `Y` = whether a patient having surgery with general anesthesia experienced a sore throat on waking (0 = no, 1 = yes) as a function of the `D` = duration of the surgery (in minutes) and the `T` = type of device used to secure the airway (0 = laryngeal mask airway, 1 = tracheal tube). Use a model-building strategy to select a logistic model for these predictors. For your model, interpret parameter estimates, and conduct inference about the effects.

Patient	*D*	*T*	*Y*	Patient	*D*	*T*	*Y*	Patient	*D*	*T*	*Y*
1	45	0	0	13	50	1	0	25	20	1	0
2	15	0	0	14	75	1	1	26	45	0	1
3	40	0	1	15	30	0	0	27	15	1	0
4	83	1	1	16	25	0	1	28	25	0	1
5	90	1	1	17	20	1	0	29	15	1	0
6	25	1	1	18	60	1	1	30	30	0	1
7	35	0	1	19	70	1	1	31	40	0	1
8	65	0	1	20	30	0	1	32	15	1	0
9	95	0	1	21	60	0	1	33	135	1	1
10	35	0	1	22	61	0	0	34	20	1	0
11	75	0	1	23	65	0	1	35	40	1	0
12	45	1	1	24	15	1	0				

Source: Data from "Binary Data" by D. Collett, in *Encyclopedia of Biostatistics*, 2nd ed. Hoboken, NJ: Wiley, 2005, pp. 439–446.

This time we try doing forward selection.

```
null_model <- glm(SoreThroat ~ 1, data = sorethroat_dataset, family = binomial)

full_model <- glm(SoreThroat ~ SurgeryDuration * DeviceType,
                  data = sorethroat_dataset,
                  family = binomial(link='logit'))

forward_selection_model <- step(
  null_model, scope = list(lower = null_model, upper = full_model),
  direction = "forward", trace = 0)
```

```
Call:
glm(formula = SoreThroat ~ SurgeryDuration + DeviceType, family = binomial,
   data = sorethroat_dataset)

Coefficients:
               Estimate Std. Error z value Pr(>|z|)
(Intercept)      -1.4173     1.0946   -1.29   0.1954
SurgeryDuration   0.0687     0.0264    2.60   0.0093 **
DeviceType1      -1.6589     0.9229   -1.80   0.0722 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

   Null deviance: 46.180  on 34  degrees of freedom
Residual deviance: 30.138  on 32  degrees of freedom
AIC: 36.14

Number of Fisher Scoring iterations: 5
```

Model interpretation in terms of odds: We take `exp` of the coefficients.

- The odds of a patient experiencing a sore throat on waking is 1.07109 times the odds for someone who had undergone the operation for a duration that is one unit less.
- The odds of a patient experiencing a sore throat on waking when tracheal tube is used to secure the airway is 0.19034 times the odds for a patient for whom laryngeal mask is used.
- The intercept does not make sense as it is impossible for a surgery to have duration of zero.

Inference about the effects:

We are 90% confident that if we repeat the sampling many times, the values of the parameters will fall between the corresponding lower (LB) and upper bounds (UB) in Table 5.1. Both do not include zero indicating their significant effect on the log-odds.

```
c90 <- 1.645; se <- summary(forward_selection_model)$coefficients[, 2]
SurgeryDuration_LB <- betahat1 - c90*se['SurgeryDuration']
SurgeryDuration_UB <- betahat1 + c90*se['SurgeryDuration']
DeviceType1_LB <- betahat2 - c90*se['DeviceType1']
DeviceType1_UB <- betahat2 + c90*se['DeviceType1']
```

Table 5.1: 90% Confidence Intervals

Variable	LB	UB
D	0.02523	0.11212
T	-3.17704	-0.14085

6 Logistic Regression VI

For the horseshoe crab data, use methods discussed in our class to evaluate predictive power for logistic models that include weight and color as explanatory variables.

We consider 2 models:

- `model_horseshoecrab_score`: Model that treats the color as score
- `model_horseshoecrab_collapsed`: Model where we collapse the color categories into dark and not dark

```
horseshoecrab_dataset <- read.csv("HorseshoeCrab.csv") %>%
  dplyr::select(color, weight, indsatell)

model_horseshoecrab_score <- glm(indsatell~color + weight,
                                 data=horseshoecrab_dataset,
                                 family=binomial(link="logit"))
```

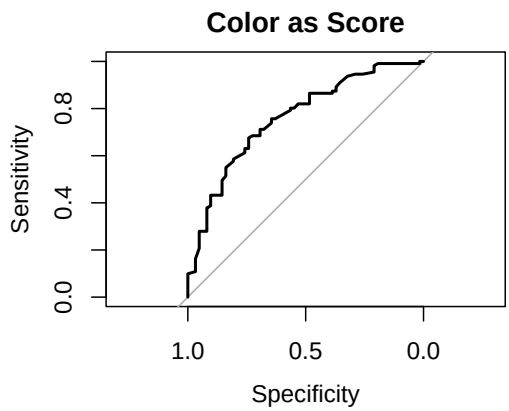
```
Call:
glm(formula = indsatell ~ color + weight, family = binomial(link = "logit"),
   data = horseshoecrab_dataset)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -1.517397   1.263965   -1.20    0.230
color       -0.514180   0.223351   -2.30    0.021 *
weight       0.001653   0.000382    4.32 0.000015 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

   Null deviance: 225.76  on 172  degrees of freedom
Residual deviance: 190.27  on 170  degrees of freedom
AIC: 196.3

Number of Fisher Scoring iterations: 4
```

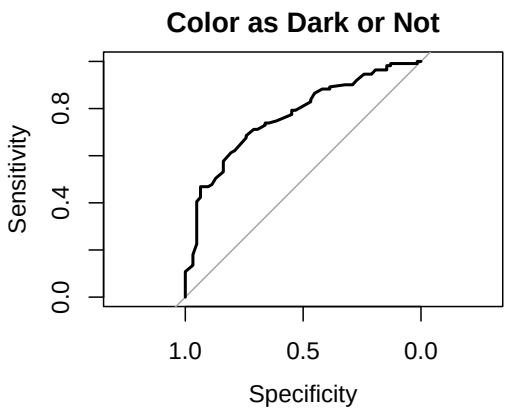
```
horseshoecrab_dataset_collapsed <- horseshoecrab_dataset %>%
  mutate(color = factor(ifelse(color %in% 2:4, 0, 1)))

model_horseshoecrab_collapsed <- glm(indsatell~color + weight,
                                     data=horseshoecrab_dataset_collapsed,
                                     family=binomial(link="logit"))
```

```
Call:
glm(formula = indsatell ~ color + weight, family = binomial(link = "logit"),
   data = horseshoecrab_dataset_collapsed)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -3.313445   0.898444   -3.69  0.00023 ***
color1      -1.295368   0.522164   -2.48  0.01311 *
weight       0.001729   0.000383    4.52  6.2e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

   Null deviance: 225.76  on 172  degrees of freedom
Residual deviance: 189.17  on 170  degrees of freedom
AIC: 195.2

Number of Fisher Scoring iterations: 4
```

```r
actual <- horseshoecrab_dataset$indsatell
Predictedcollapsed <- predict(model_horseshoecrab_collapsed, type="response")
Predictedscore <- predict(model_horseshoecrab_score, type="response")
roc_obj_collapsed <- roc(actual, Predictedcollapsed)
roc_obj_score <- roc(actual, Predictedscore)
```

The Area Under the Curve or AUC is 0.76053 for when color is treated as score and 0.76758 when color is either dark or not. This is better when higher.





Suppose we select 0.5 to be the threshold to obtain the confusion matrix.

```r
threshold <- 0.5
Predictedscore <- ifelse(predict(model_horseshoecrab_score,
                                 type="response")>threshold,1,0)
Predictedcollapsed <- ifelse(predict(model_horseshoecrab_collapsed,
                                     type="response")>threshold,1,0)
conf_matrix_score <- table(Predictedscore,actual)
conf_matrix_collapsed <- table(Predictedcollapsed,actual)
```

Table 6.1: Comparison of Confusion Matrices Across Models

Color as Score			Color as Dark or Not		
	Actual			Actual	
Pred	0	1	Pred	0	1
0	30	16	0	28	15
1	32	95	1	34	96

`caret` produces a complete list of metrics. Some of which are shown in Table 6.2.

```
actual <- factor(actual, levels = c(0, 1))
Predictedcollapsed <- factor(Predictedcollapsed, levels = c(0, 1))
Predictedscore <- factor(Predictedscore, levels = c(0, 1))

caret_score <- caret::confusionMatrix(Predictedscore, actual,
                                      mode = "everything", positive = "1")
caret_collapsed <- caret::confusionMatrix(Predictedcollapsed, actual,
                                          mode = "everything", positive = "1")
```

Table 6.2: Predictive Power Metrics

Model Description	Sensitivity	Specificity	Precision	Recall	F1	Accuracy
Color as Score	0.856	0.484	0.748	0.856	0.798	0.723
Color as Dark or Not	0.865	0.452	0.738	0.865	0.797	0.717

7 Appendix

Interpreting 2-way interaction model coefficients using Item 5.

```
Call:
glm(formula = SoreThroat ~ SurgeryDuration * DeviceType, family = binomial(link = "logit"),
   data = sorethroat_dataset)

Coefficients:
                          Estimate Std. Error z value Pr(>|z|)
(Intercept)                 0.0498     1.4694    0.03     0.97
SurgeryDuration             0.0285     0.0343    0.83     0.41
DeviceType1                -4.4722     2.4671   -1.81     0.07 .
SurgeryDuration:DeviceType1  0.0746     0.0578    1.29     0.20
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

   Null deviance: 46.180  on 34  degrees of freedom
Residual deviance: 28.321  on 31  degrees of freedom
AIC: 36.32

Number of Fisher Scoring iterations: 6
```

$$\frac{\hat{\pi}(x)}{1-\hat{\pi}(x)} = \exp(\hat{\beta}_0 + \hat{\beta}_1 D + \hat{\beta}_2 T + \hat{\beta}_3 DT)$$

When using tracheal tube:

$$\begin{aligned}
\frac{\hat{\pi}(x+1)}{1-\hat{\pi}(x+1)} &= \exp(\hat{\beta}_0 + \hat{\beta}_1 D + \hat{\beta}_2 (T+1) + \hat{\beta}_3 D(T+1)) \\
&= \exp(\hat{\beta}_0 + \hat{\beta}_1 D + \hat{\beta}_2 T + \hat{\beta}_2 + \hat{\beta}_3 DT + \hat{\beta}_3 D) \\
&= \exp(\hat{\beta}_0 + \hat{\beta}_1 D + \hat{\beta}_2 T + \hat{\beta}_3 DT) \exp(\hat{\beta}_2 + \hat{\beta}_3 D) \\
&= \frac{\hat{\pi}(x)}{1-\hat{\pi}(x)} \exp(\hat{\beta}_2 + \hat{\beta}_3 D) \\
&= \frac{\hat{\pi}(x)}{1-\hat{\pi}(x)} \exp(\hat{\beta}_2) \exp(\hat{\beta}_3 D)
\end{aligned}$$

When duration is increased by one unit:

$$\begin{aligned}
\frac{\hat{\pi}(x+1)}{1-\hat{\pi}(x+1)} &= \exp(\hat{\beta}_0 + \hat{\beta}_1 (D+1) + \hat{\beta}_2 T + \hat{\beta}_3 T(D+1) \\
&= \exp(\hat{\beta}_0 + \hat{\beta}_1 D + \hat{\beta}_1 + \hat{\beta}_2 T + \hat{\beta}_3 DT + \hat{\beta}_3 T) \\
&= \exp(\hat{\beta}_0 + \hat{\beta}_1 D + \hat{\beta}_2 T + \hat{\beta}_3 DT) \exp(\hat{\beta}_1 + \hat{\beta}_3 T) \\
&= \frac{\hat{\pi}(x)}{1-\hat{\pi}(x)} \exp(\hat{\beta}_1 + \hat{\beta}_3 T) \\
&= \frac{\hat{\pi}(x)}{1-\hat{\pi}(x)} \exp(\hat{\beta}_1) \exp(\hat{\beta}_3 T) \\
&\to \frac{\hat{\pi}(x)}{1-\hat{\pi}(x)} \exp(\hat{\beta}_1) \exp(\hat{\beta}_3) \qquad \text{when T} = 1 \\
&\to \frac{\hat{\pi}(x)}{1-\hat{\pi}(x)} \exp(\hat{\beta}_1) \qquad \text{when T} = 0
\end{aligned}$$

```r
betahat1 <- full_model$coefficients["SurgeryDuration"]
betahat2 <- full_model$coefficients["DeviceType1"]
betahat3<- full_model$coefficients["SurgeryDuration:DeviceType1"]
```

Model interpretation:

- (log odds - uses β's) For those that use laryngeal mask airway device type, the log odds of experiencing sore throat upon waking increases by 0.02848 for every unit increase in surgery duration but increases by 0.10308 for those that use tracheal tube. (odds - uses $\exp(\beta)$'s) The odds of experiencing sore throat upon waking when using tracheal tube is 1.10858 times the odds when surgery is one unit shorter in duration while the odds of experiencing sore throat upon waking when using laryngeal mask airway is 1.02889 times the odds when surgery is one unit shorter in duration.
- (odds) The odds of experiencing sore throat upon waking when using tracheal tube is $0.01142 \times \exp(\ 0.0746\ \times D)$ the odds when using laryngeal mask airway device.
- The intercept does not make sense as it is impossible for a surgery to have duration of zero.